

WHAT DOES A PROMPT COST?

An Interactive Experiment on the Environmental Cost of AI

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FORMAT Interactive Digital Poster / Web Experience

TECHNOLOGY HTML · CSS · JavaScript · Node.js · Express · Google Gemini API

1. PROJECT CONCEPT

"What Does a Prompt Cost?" is an interactive digital poster exploring the hidden physical and environmental footprint behind everyday AI interactions. A visitor types a question, receives an AI-generated response, and immediately sees estimated energy, heat, and water-footprint values. A running session total makes the cumulative effect visible.

2. THE EXPERIENCE

The interaction follows: **PROMPT** → **COMPUTATION** → **ENERGY** → **HEAT** → **COOLING** → **WATER FOOTPRINT**. Environmental information appears directly after the visitor's own interaction, connecting an ordinary question with an otherwise invisible computational process.

3. ENVIRONMENTAL COST

The project uses a published reference value of approximately **0.24 Wh of energy** and **0.26 mL of water** as a median reference for a text-based AI prompt. Heat is represented as an energy equivalent rather than a temperature increase. Using **1 Wh = 3.6 kJ**, 0.24 Wh corresponds to approximately **0.86 kJ**.

4. ESTIMATION METHODOLOGY

The live application receives token-usage information from the AI API and uses it to scale the reference environmental values according to the relative computational size of each interaction. These are **estimates, not direct measurements**. The interface states: "**ESTIMATED VALUES · ACTUAL IMPACT VARIES.**"

5. IMPORTANT LIMITATIONS

There is no single universal environmental cost for every AI prompt. Actual impact can vary with the AI model, input and output tokens, hardware, workload, data-centre efficiency, cooling system, location, electricity source, and infrastructure utilisation. The displayed values are intended as an educational approximation.

6. DESIGN APPROACH

The visual language is based on a familiar AI chat interface. Environmental information appears immediately after the visitor's interaction, creating a direct relationship between **QUESTION** → **ANSWER** → **COST**.

7. TECHNICAL IMPLEMENTATION

Frontend: HTML, CSS, JavaScript

Backend: Node.js, Express

AI: Google Gemini API

The API key is stored as a server-side environment variable and is not exposed in the frontend or GitHub repository.

8. PROJECT STRUCTURE

public/ — app.js · index.html · style.css

server.js · package.json · package-lock.json · .gitignore

9. LIVE PROJECT & SOURCE

LIVE PROJECT <https://what-does-a-prompt-cost.onrender.com/>

GITHUB <https://github.com/swayam0333777-stack/what-does-a-prompt-cost>

10. PROJECT INTENTION

The project asks: **If asking an AI a question feels almost weightless, what does that question actually cost?** By turning an invisible computational process into a visible physical estimate, the project encourages visitors to think about the infrastructure, energy, and resources behind seemingly effortless digital interactions.

ONE QUESTION IS TINY. MILLIONS OF QUESTIONS AREN'T.